JOC The Journal of Organic Chemistry

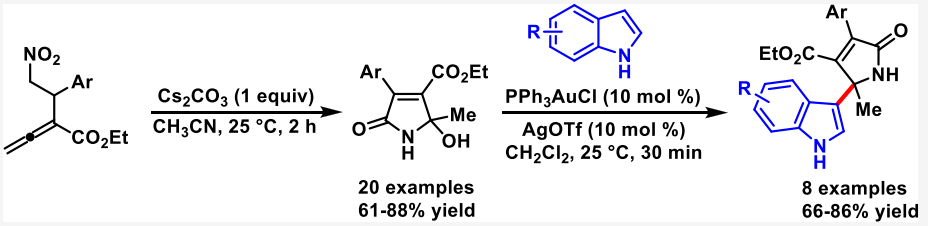
pubs.acs.org/joc Article

Base-Mediated Intramolecular Cyclization of α-Nitroethylallenic Esters as a Synthetic Route to 5-Hydroxy-3-pyrrolin-2-ones

Narendra Kumar Vaishanv, Mohd Khalid Zaheer, Sandeep Kumar, Ruchir Kant, and Kishor Mohanan*

Cite This: J. Org. Chem. 2021, 86, 5630−5638 Read Online

ACCESS | Metrics & More | Article Recommendations | Supporting Information



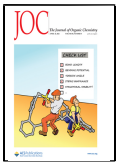
ABSTRACT: An unprecedented Cs_2CO_3-mediated intramolecular cyclization/rearrangement cascade that transforms α-nitroethylallenic esters to functionalized pyrrolin-2-ones has been uncovered. This reaction provides a new and practical approach for the synthesis of medicinally privileged 5-hydroxy-3-pyrrolin-2-ones under mild conditions. The broad potential of this new method was demonstrated by an efficient Au/Ag-catalyzed heteroarylation of 5-hydroxy-3-pyrrolin-2-ones employing electron-rich heteroarenes to furnish heteroaryl-lactam derivatives.

Downloaded via FUDAN UNIV on January 18, 2022 at 02:41:28 (UTC).
See https://pubs.acs.org/sharingguidelines for options on how to legitimately share published articles.

The 5-hydroxy-3-pyrrolin-2-one is an important class of heterocyclic compounds frequently encountered in biologically active natural products.[1] Epolactaene, oteromycin, and pyrrocidine A are some of the selected examples for the bioactive natural products containing the hydroxypyrrolinone motif which display neuritogenic as well as other pharmacological properties (Figure 1).[2] Furthermore, 5-hydroxy-3-pyrrolin-2-ones have also been employed as a critical intermediate in the synthesis of various natural products.[3] Consequently, over the past decades, several methods have been developed to access this important class of compounds. For instance, intramolecular cyclization reactions, intermolecular reactions, and oxidation of heteroaryls have enabled facile access to functionalized 5-hydroxy-3-pyrrolin-2-ones.[4] Among these methods, intramolecular reactions involving a tethered amide moiety is the most commonly used approach, and some selected examples include (i) Cu-catalyzed haloacetamization of allenamides, (ii) halolactamization of homoallylic amides, and (iii) the intramolecular radical/enaminic cyclization reactions of prefunctionalized substrates such as enamides (Scheme 1a,b).[5,6] Intermolecular reactions have also provided straightforward methods to access 5-hydroxy-3-pyrrolin-2-ones. The reaction between nitroolefins and 1,3-diketones[7] and the nucleophilic addition/cyclization of amines to MBH adducts constitute mild routes to hydroxypyrrolinones (Scheme 1c).[8] However, the development of novel protocols to access this class of compounds is always in high demand.

Activated allenes have emerged as a valuable synthon in organic transformations.[9] In particular, nucleophilic phosphine-catalyzed annulation reactions involving allenic esters have been extensively explored for the construction of a variety of carbo- and heterocycles.[10] Besides nucleophilic phosphine catalysis, the synthesis of heterocycles through the base-mediated reaction of various nucleophiles or electrophiles to allenic esters have also been explored, though to a limited extent.[11] In 2006, the Shi group disclosed a carbonate-catalyzed cyclization reaction of salicylic aldehydes with allenic ester to obtain chromene derivatives in good to excellent yield.[12] Subsequently, Selig, Zhou, and Rouh demonstrated the utility of base-mediated reactions of allenic esters in constructing various heterocyclic compounds such as quinolines, benzooxazepino[5,4-*a*]isoindolone, and functionalized chromene and chroman derivatives.[13−15] The Kwon group recently disclosed an interesting phosphine-catalyzed intramolecular cyclization of α-nitroethylallenic esters to access five-membered cyclic *N*-hydroxyimidic acid esters.[16] However, it remained as the only report on the reactivity of α-nitroethylallenic esters, and related cyclization reactions under basic conditions are unknown. Our investigations in this direction culminated in a straightforward single-step synthesis of 5-hydroxy-3-pyrrolin-2-ones via a Cs_2CO_3-mediated intramolecular cyclization of α-nitroethylallenic esters (Scheme 1d).

Received: January 14, 2021
Published: March 31, 2021



ACS Publications © 2021 American Chemical Society

https://doi.org/10.1021/acs.joc.1c00109
J. Org. Chem. 2021, 86, 5630−5638

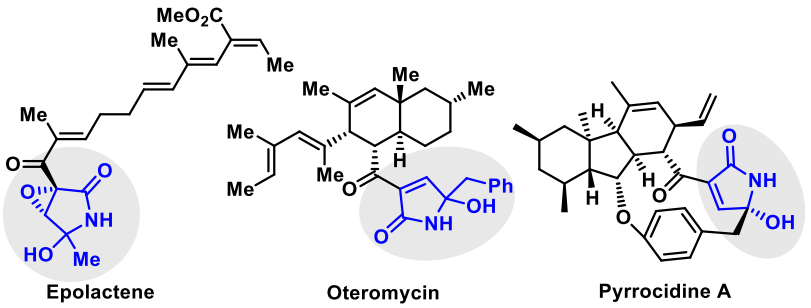
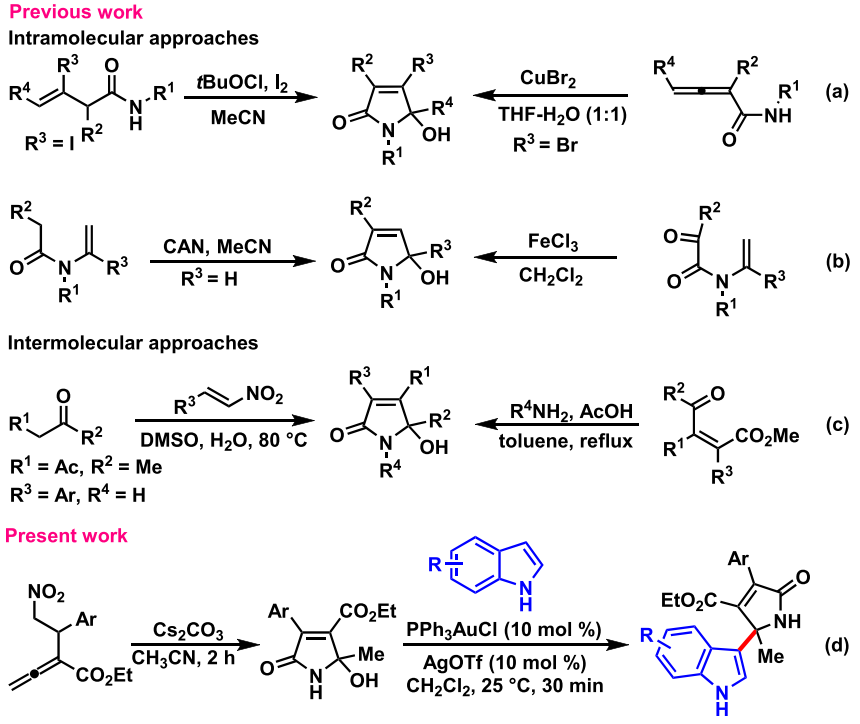


Figure 1. 5-Hydroxypyrrolin-2-one motif in bioactive natural products.

Scheme 1. Synthetic Routes to 5-Hydroxy-3-pyrrolin-2-ones

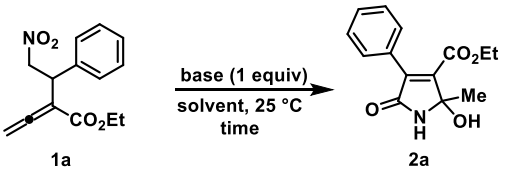


With these precedents in hand, we commenced our study by examining the reaction of ethyl 2-(2-nitro-1-phenylethyl) buta-2,3-dienoate **1a** with CsF in CH_3CN at room temperature. To our surprise, the substrate underwent an unexpected intramolecular cascade reaction to form 5-hydroxy-3-pyrrolin-2-one **2a** in 66% yield (Table 1, entry 1). This unprecedented result prompted us to screen the reaction parameters in greater detail to improve the reaction efficiency. Preliminary solvent screening indicated that CH_3CN was optimal among a variety of solvents (entries 1−8). Subsequently, the influence of various organic and inorganic bases in CH_3CN was investigated. The use of inorganic bases such as *t*-BuOK, KOH, NaOH, and K_2CO_3 provided the product in moderate yields, while Cs_2CO_3 gave a better yield, in short reaction time (entries 9−13). There was no improvement in the efficiency when the reaction was tried using organic bases such as DMAP and DABCO (entries 14,15). The product **2a** was obtained only in low yield when we attempted to render the reaction substoichiometric by employing 0.5 equiv of Cs_2CO_3 (entry 16). Moreover, no product yield enhancement was observed when the reaction was carried out at a higher temperature.

Having the optimal conditions identified, we set out to assess the generality of the present method with respect to various substitution patterns on the α-nitroethylallenic esters for the construction of hydroxypyrrolinones, and the results are summarized in Scheme 2. Under the standard conditions, a variety of nitroethylallenic esters, irrespective of the electronic and steric nature of the substituents they possess, were converted to the corresponding products in good yields. Aromatic rings bearing electronically varied substituents such as methoxy, thiomethyl, bromo, chloro, and fluoro at the *para*-position of the phenyl ring were very well tolerated and provided the desired products in good to excellent yields (**2b**−**h**). The structure of the product **2f** was unambiguously confirmed by X-ray analysis. Delightfully, the transformation was effective for substrates bearing methoxy as well as halo substituents at the *meta*-position (**2i**−**k**). Furthermore, the substrates having di- and trisubstituted aromatic rings were also competent in this reaction, providing the corresponding products without any deterioration in yield (**2l**−**m**). The protocol was also amenable for naphthyl-derived substrates and furnished the product **2n** in 88% yield. Given the ubiquity of heteroaryls in drug discovery, we evaluated substrates bearing thiophene and furan for this reaction, and the corresponding hydroxypyrrolinones were achieved in good yields (**2o**−**q**). Finally, we sought to evaluate the substrate

https://doi.org/10.1021/acs.joc.1c00109
J. Org. Chem. 2021, 86, 5630−5638

Table 1. Optimization of the Reaction Conditions[a]



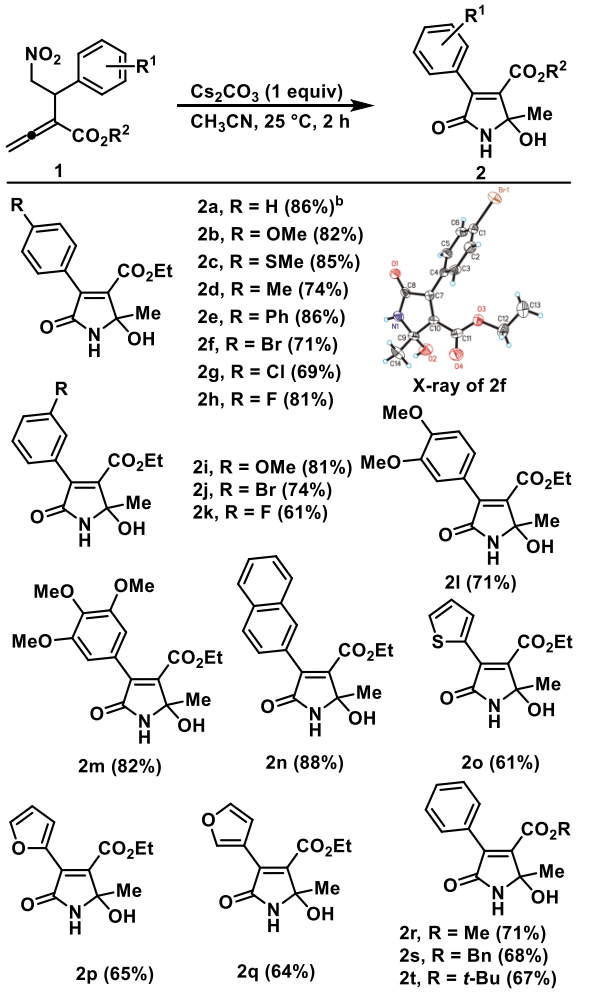
entry	base	solvent	time (h)	yield of 2a[b] (%)
1	CsF	CH_3CN	12	66
2	CsF	CH_2Cl_2	12	15
3	CsF	THF	12	<5
4	CsF	$CHCl_3$	12	<5
5	CsF	EtOAc	12	11
6	CsF	MeOH	12	<5
7	CsF	DMF	12	16
8	CsF	DMSO	12	12
9	*t*-BuOK	CH_3CN	12	15
10	NaOH	CH_3CN	2	40
11	KOH	CH_3CN	2	65
12	K_2CO_3	CH_3CN	2	61
13	Cs_2CO_3	CH_3CN	2	86
14	DMAP	CH_3CN	12	55
15	DABCO	CH_3CN	12	75
16[c]	Cs_2CO_3	CH_3CN	12	66
17[d]	Cs_2CO_3	CH_3CN	2	85

[a]General conditions: **1a** (0.20 mmol), base (0.20 mmol), and solvent (2.0 mL). [b]Isolated yield after silica gel column chromatography. [c]0.10 mmol of base was used. [d]Reaction carried out at 70 °C.

scope by varying the ester moiety of allene **1**, and the methyl, benzyl, and *t*-butyl esters of allenoate readily underwent the reaction to furnish the products in good yields (**2r**–**t**). Our further attempts to expand the scope of allenoates by introducing substitution at the γ-position failed to deliver the anticipated product. On a larger scale, the reaction of nitroethylallenoate **1a** provided the corresponding product **2a** in 66% yield.

Next, we focused our attention on further functionalization of 5-hydroxy-3-pyrrolin-2-one **2**. We anticipated that a π acid such as PPh_3AuCl might coordinate with the hydroxyl group of pyrrolinone moiety **2a** and generate the *N*-acyliminium ion, making it an effective electrophile for electron-rich heteroaryls. Given the importance of indoles as a common motif in natural products and bioactive compounds,[17] we selected 1-methylindole as a model substrate for this reaction. If successful, this reaction would provide a practical route to indolylpyrrolinones. Based on these considerations, the reaction of **2a** with 1-methylindole was attempted with catalytic amounts of PPh_3AuCl and AgOTf in DCM. To our delight, the reaction proceeded in the expected manner, and indolylpyrrolinone **4a** was obtained in 86% yield (Scheme 3). Notably, N-unsubstituted indole reacted analogously to afford product **4b** in 77% yield. Moreover, halo substituents at the C5- or C6-position of the indole ring were tolerated, affording the corresponding products in good yields (**4c**, **4d**). A bulkier tetrahydrocyclopentaindole substrate also underwent this reaction and provided **4e** in high yield. The structure of **4e** was confirmed by X-ray crystallographic analysis. The reaction carried out using 3-methylindole furnished the C2 product **4f** in good yield, illustrating the applicability of this method for structurally challenging substrates. Other heteroaryl substrates such as 2-methoxythiophene and 1-methylpyrrole also reacted

Scheme 2. Synthesis of 5-Hydroxy-3-pyrrolin-2-ones: Scope of α-(Nitroethyl)allenic Esters[a]



[a]General conditions: **1** (0.20 mmol), Cs_2CO_3 (0.20 mmol), CH_3CN (2.0 mL), 25 °C, 2 h. Isolated yield after silica gel column chromatography. [b]The reaction on a 1.0 g scale afforded **2a** in 66% yield.

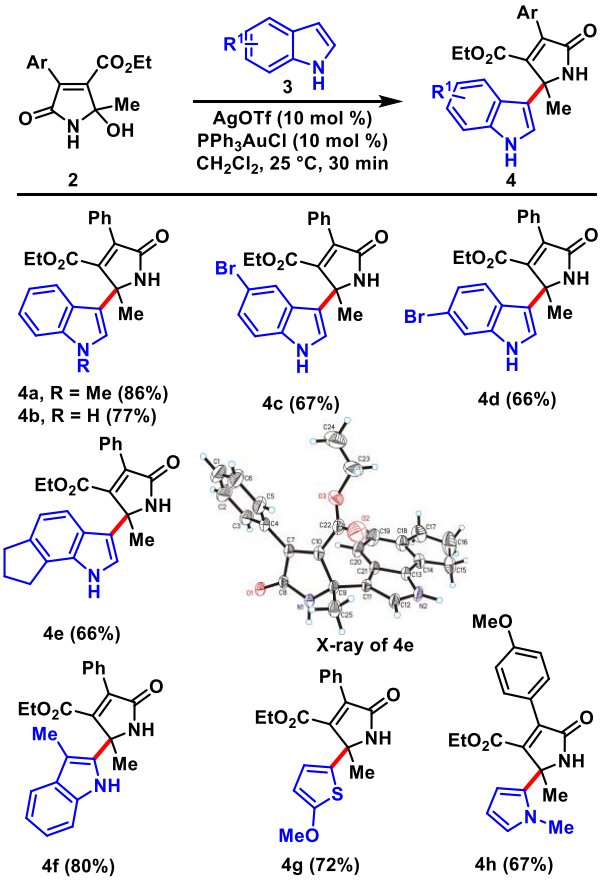
smoothly with hydroxypyrrolinone and delivered the products in good yields (**4g**, **4h**).

Besides this, in order to leverage the synthetic utility of the products in accessing bis-indole alkaloid-like compounds,[18] the lactam **4b**, after the N-Boc protection of both indole and pyrrole moieties, was subjected to reduction using excess DIBAL-H. The reduction delivered the hemi aminal **6** as an inseparable mixture of diastereomers in 5:1 ratio. The product **6** was then treated with *N*-methylindole, and pleasingly, the reaction furnished the bisindole derivative **7** in 67% yield, albeit with low diastereoselectivity (Scheme 4). To further demonstrate the synthetic relevance of the method, reductive removal of the hydroxyl group of **2a** was performed using $NaCNBH_3$, and the corresponding lactam **8** was obtained in 56% yield. Treating **2a** with HCl and methanol delivered the methyl ether **9** in 70% yield.

A plausible mechanism for the formation of 5-hydroxy-3-pyrrolin-2-ones is outlined in Scheme 5. First, carbonate-mediated deprotonation of α-nitroethylallenic ester **1** forms the corresponding α-nitro anion **A**. The alternative resonance form of anion **A**, nitronate **B**, engages in intramolecular 1,4-

https://doi.org/10.1021/acs.joc.1c00109
J. Org. Chem. 2021, 86, 5630–5638

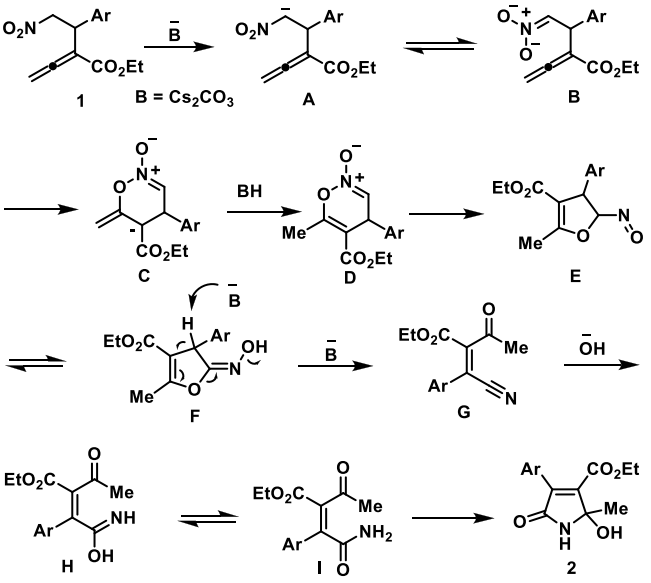
Scheme 3. Heteroarylation of 5-Hydroxy-3-pyrrolin-2-ones



[a]General conditions: **2** (0.20 mmol), **3** (0.26 mmol), PPh_3AuCl (0.02 mmol), AgOTf (0.02 mmol), CH_2Cl_2 (2.0 mL), 25 °C, 30 min. Isolated yield after silica gel column chromatography.

conjugate addition to generate intermediate **C**, which subsequently isomerizes to give the cyclic nitronate **D**. The

Scheme 5. Proposed Mechanistic Pathway

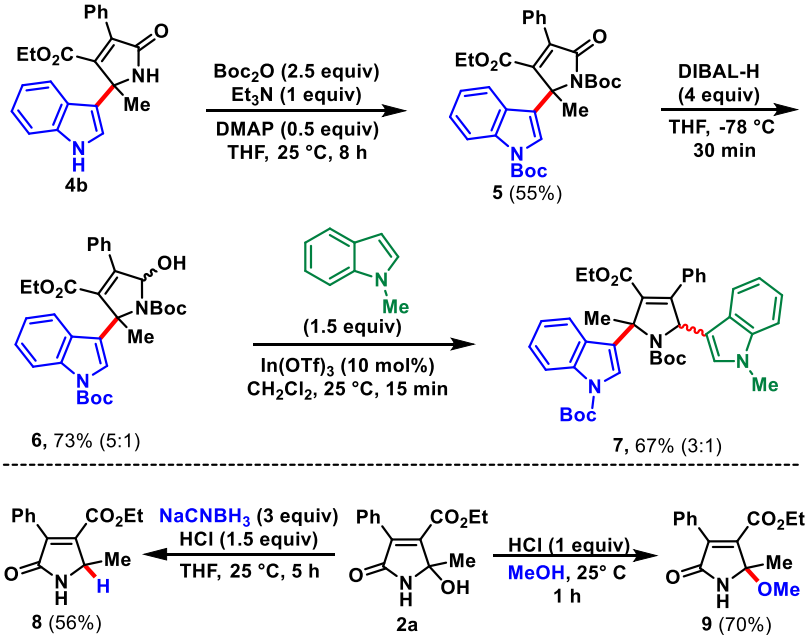


nitronate **D** involves a spontaneous ring contraction/tautomerization process to form the oxime intermediate **F**.[16] Oxime **F**, under the basic conditions, undergoes ring opening to give rise to the nitrile intermediate **G**. Subsequent hydrolysis of **G** affords the intermediate **H**, which tautomerizes to amide **I**. Finally, the intramolecular cyclization of **I** affords the desired 5-hydroxy-3-pyrrolin-2-ones **2**.

To shed light into the proposed reaction mechanism, the oxime intermediate **F** was synthesized using a reported procedure.[16] Pleasingly, the substrate **F**, when exposed to Cs_2CO_3 under the established reaction conditions, readily transformed into compound **2a** in 87% yield. This ascertains the intermediacy of **F** in this novel protocol (Scheme 6).

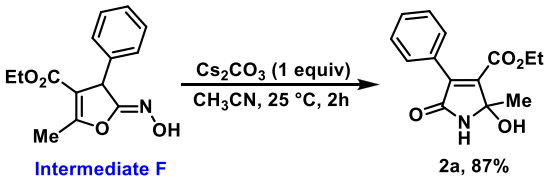
In summary, we have reported an efficient base-mediated intramolecular transformation of α-nitroethylallenic esters to

Scheme 4. Synthetic Modifications



https://doi.org/10.1021/acs.joc.1c00109
J. Org. Chem. 2021, 86, 5630−5638

Scheme 6. Mechanistic Investigation



functionalized pyrrolin-2-ones. This protocol offers access to a wide range of medicinally valuable 5-hydroxy-3-pyrrolin-2-ones in good to excellent yield and features a wide substrate scope and high functional group tolerance. The utility of this method was subsequently demonstrated by developing an Au/Ag-catalyzed heteroarylation of 5-hydroxy-3-pyrrolin-2-one to access densely functionalized heteroaryl-lactam derivatives. The strategy was subsequently extended to access synthetically attractive bis-indole-pyrroline derivatives.

EXPERIMENTAL SECTION

General Experimental Information. Unless otherwise specified, all reactions were performed under air atmosphere in oven-dried round-bottom flasks. The reactions were monitored by TLC visualized by UV (254 nm) and/or with iodine. Column chromatography was performed on 100−200 mesh silica gel using the gradient system ethyl acetate-hexane. The NMR data were recorded at Bruker AV 400 MHz in $CDCl_3$/DMSO-d_6 using as internal standards the residual $CHCl_3$ signal for ^1H NMR (δ = 7.26 ppm) and the deuterated solvent signal for ^{13}C NMR (δ = 77.16 ppm) and residual DMSO signal for ^1H NMR (δ = 2.50 ppm) and the deuterated solvent signal for ^{13}C NMR (δ = 39.5 ppm). Coupling constants are given in Hertz (Hz), and the classical abbreviations are used to describe the signal multiplicities. Melting points were measured with a Büchi B-540 apparatus and are uncorrected. High-resolution mass spectra were obtained using a Q-TOF mass spectrometer. All commercially available reagents were used as received. α-Nitroethylallenic esters **1a−t** were synthesized and characterized by following the procedure reported by the Kwon group.[16]

General Procedure for the Synthesis of 5-Hydroxy-3-pyrrolin-2-ones 2a−t. To an oven-dried round-bottom flask was added α-nitroethylallenic ester **1** (0.20 mmol) in CH_3CN and Cs_2CO_3 (0.20 mmol). The reaction mixture was stirred at 25 °C for 2 h. After the completion of the reaction, as indicated by TLC, the reaction mixture was extracted using ethyl acetate (3 × 10 mL) and water, and the organic layer was washed with saturated brine. Finally, the organic layer was dried over anhydrous sodium sulfate and evaporated under reduced pressure. The residue was purified using column chromatography (100−200 mesh silica gel) using acetone/dichloromethane as the eluent to afford the product **2**.

Procedure for the Gram-Scale Synthesis of 5-Hydroxy-3-pyrrolin-2-ones 2a. To an oven-dried round-bottom flask was added α-nitroethylallenic ester **1a** (3.82 mmol, 1.0 g) in 5 mL of CH_3CN and Cs_2CO_3 (3.82 mmol, 1.247 g). The reaction mixture was stirred at 25 °C for 2 h. After the completion of reaction, as indicated by TLC, the reaction mixture was extracted using ethyl acetate (3 × 20 mL) and water and was washed with saturated brine. Finally, the organic layer was dried over anhydrous sodium sulfate and evaporated under reduced pressure. The residue was purified using column chromatography (100−200 mesh silica gel) using acetone/dichloromethane as the eluent to afford the product **2a** in 66% yield (660 mg).

*Ethyl 2-Hydroxy-2-methyl-5-oxo-4-phenyl-2,5-dihydro-1H-pyrrole-3-carboxylate (**2a**).* Yellow solid, 45 mg, 86%; R_f(acetone/DCM: 2/8) = 0.47. Mp 150−152 °C. $^{13}C\{^1H\}$ NMR (100 MHz, δ ppm/$CDCl_3$): 169.2, 163.8, 144.1, 139.5, 129.7, 129.3, 128.0, 85.7, 61.8, 25.5, 13.9. ^1H NMR (400 MHz, δ ppm/$CDCl_3$): 7.48−7.38 (m, 5H), 7.19 (s, 1H), 4.26−4.20 (m, 2H), 3.97 (s, 1H), 1.79 (d, J = 2.0 Hz, 3H), 1.16 (td, J = 7.2, 1.6 Hz, 3H). HRMS (ESI) m/z: $[M + H]^+$ calcd for $C_{14}H_{16}NO_4$ 262.1074; found 262.1075.

*Ethyl 2-Hydroxy-4-(4-methoxyphenyl)-2-methyl-5-oxo-2,5-dihydro-1H-pyrrole-3-carboxylate (**2b**).* White solid, 48 mg, 82%; R_f (acetone/DCM: 2/8) = 0.41. Mp 152−154 °C. $^{13}C\{^1H\}$ NMR (100 MHz, δ ppm/$CDCl_3$): 169.4, 164.1, 161.0, 142.0, 139.1, 131.6, 121.5, 113.5, 85.5, 61.8, 55.4, 25.7, 14.0. ^1H NMR (400 MHz, δ ppm/$CDCl_3$): 7.50−7.47 (m, 2H), 6.97 (s, 1H), 6.92−6.89 (m, 2H), 4.26 (q, J = 7.2 Hz, 2H), 3.88 (s, 1H), 3.83 (s, 3H), 1.79 (s, 3H), 1.22 (t, J = 7.2 Hz, 3H). HRMS (ESI) m/z: $[M + H]^+$ calcd for $C_{15}H_{18}NO_5$ 292.1179; found 292.1183.

*Ethyl 2-Hydroxy-2-methyl-4-(4-(methylthio)phenyl)-5-oxo-2,5-dihydro-1H-pyrrole-3-carboxylate (**2c**).* Red solid, 52 mg, 85%; R_f (acetone/DCM: 2/8) = 0.43. Mp 128−130 °C. $^{13}C\{^1H\}$ NMR (100 MHz, δ ppm/$CDCl_3$): 169.2, 163.9, 143.1, 141.3, 138.8, 130.2, 125.5, 125.3, 125.3, 85.7, 61.8, 25.5, 15.3, 14.0. ^1H NMR (400 MHz, δ ppm/$CDCl_3$): 7.44−7.42 (m, 2H), 7.24−7.21 (m, 2H), 7.05 (s, 1H), 4.26 (q, J = 7.2 Hz, 2H), 3.85 (s, 1H), 2.49 (s, 3H), 1.79 (s, 3H), 1.21 (t, J = 7.0 Hz, 3H). HRMS (ESI) m/z: $[M + H]^+$ calcd for $C_{15}H_{18}NO_4S$ 308.0951; found 308.0944.

*Ethyl 2-Hydroxy-2-methyl-5-oxo-4-(p-tolyl)-2,5-dihydro-1H-pyrrole-3-carboxylate (**2d**).* White solid, 41 mg, 74%; R_f (acetone/DCM: 2/8) = 0.45. Mp 150−152 °C. $^{13}C\{^1H\}$ NMR (100 MHz, δ ppm/$CDCl_3$): 169.1, 164.0, 142.9, 140.1, 139.6, 129.8, 129.8, 128.8, 128.8, 126.2, 85.5, 61.8, 25.5, 21.6, 14.0. ^1H NMR (400 MHz, δ ppm/$CDCl_3$): 7.40 (d, J = 8.0 Hz, 2H), 7.20 (d, J = 8.0 Hz, 2H), 6.71 (s, 1H), 4.26 (q, J = 7.2 Hz, 2H), 3.58 (s, 1H), 2.38 (s, 3H), 1.81 (s, 3H), 1.21 (t, J = 7.0 Hz, 3H). HRMS (ESI) m/z: $[M + H]^+$ calcd for $C_{15}H_{18}NO_4$ 276.1230; found 276.1241.

*Ethyl 4-([1,1′-Biphenyl]-4-yl)-2-hydroxy-2-methyl-5-oxo-2,5-dihydro-1H-pyrrole-3-carboxylate (**2e**).* Orange solid, 58 mg, 86%; R_f (EtOAc/hexane: 3/7) = 0.45. Mp 115−117 °C. $^{13}C\{^1H\}$ NMR (100 MHz, δ ppm/$CDCl_3$): 169.1, 163.9, 143.6, 142.6, 140.6, 139.3, 130.3, 129.0, 128.1, 127.9, 127.3, 127.3, 126.7, 85.7, 61.9, 25.7, 14.0. ^1H NMR (400 MHz, δ ppm/$CDCl_3$): 7.64−7.58 (m, 6H), 7.46−7.43 (m, 2H), 7.39−7.35 (m, 1H), 6.90 (s, 1H), 4.28 (q, J = 7.2 Hz, 2H), 1.84 (s, 3H), 1.21 (t, J = 7.0 Hz, 3H). HRMS (ESI) m/z: $[M + H]^+$ calcd for $C_{20}H_{20}NO_4$ 338.1387; found 338.1390.

*Ethyl 4-(4-Bromophenyl)-2-hydroxy-2-methyl-5-oxo-2,5-dihydro-1H-pyrrole-3-carboxylate (**2f**).* Orange solid, 48 mg, 71%; R_f (acetone/DCM: 2/8) = 0.47. Mp 88−90 °C. $^{13}C\{^1H\}$ NMR (100 MHz, δ ppm/$CDCl_3$): 168.8, 163.5, 144.2, 138.6, 131.4, 131.3, 128.0, 124.4, 85.8, 62.0, 25.4, 14.0. ^1H NMR (400 MHz, δ ppm/$CDCl_3$): 7.51 (d, J = 5.2 Hz, 2H), 7.35 (d, J = 5.2 Hz, 2H), 7.10 (s, 1H), 4.25 (q, J = 7.2 Hz, 2H), 3.92 (s, 1H), 1.79 (s, 3H), 1.20 (t, J = 10.8 Hz, 3H). HRMS (ESI) m/z: $[M + H]^+$ calcd for $C_{14}H_{15}BrNO_4$ 340.0179; found 340.0179.

*Ethyl 4-(4-Chlorophenyl)-2-hydroxy-2-methyl-5-oxo-2,5-dihydro-1H-pyrrole-3-carboxylate (**2g**).* White solid, 41 mg, 69%; R_f (EtOAc/hexane: 3/7) = 0.47. Mp 123−125 °C. $^{13}C\{^1H\}$ NMR (100 MHz, δ ppm/$CDCl_3$): 169.0, 163.5, 144.1, 138.5, 136.1, 131.2, 128.3, 127.6, 85.7, 62.0, 25.5, 14.0. ^1H NMR (400 MHz, δ ppm/$CDCl_3$): 7.44−7.42 (m, 2H), 7.37−7.35 (m, 2H), 7.04 (s, 1H), 4.26 (q, J = 7.2 Hz, 2H), 3.84 (s, 1H), 1.80 (s, 3H), 1.20 (t, J = 7.2 Hz, 3H). HRMS (ESI) m/z: $[M + H]^+$ calcd for $C_{14}H_{15}ClNO_4$ 296.0684; found 296.0689.

*Ethyl 4-(4-Fluorophenyl)-2-hydroxy-2-methyl-5-oxo-2,5-dihydro-1H-pyrrole-3-carboxylate (**2h**).* Yellow solid, 45 mg, 81%; R_f (acetone/DCM: 2/8) = 0.46. Mp 98−100 °C. $^{13}C\{^1H\}$ NMR (100 MHz, δ ppm/$CDCl_3$): 169.2, 163.7 (C−F, $^1J_{C-F}$ = 248.0 Hz), 163.6, 143.9, 138.5, 131.9 (C−F, $^3J_{C-F}$ = 8.4 Hz,), 131.9 (C−F, $^3J_{C-F}$ = 8.4 Hz,), 125.2, 115.1 (C−F, $^2J_{C-F}$ = 21.6 Hz,), 115.1 (C−F, $^2J_{C-F}$ = 21.6 Hz,), 85.8, 61.9, 25.4, 13.9. ^1H NMR (400 MHz, δ ppm/$CDCl_3$): 7.49−7.45 (m, 2H), 7.35 (s, 1H), 7.05 (t, J = 8.8 Hz, 2H), 4.24 (q, J = 7.2 Hz, 2H), 4.16 (s, 1H), 1.78 (s, 3H), 1.19 (t, J = 7.0 Hz, 3H). HRMS (ESI) m/z: $[M + H]^+$ calcd for $C_{14}H_{15}FNO_4$ 280.0980; found 280.0988.

*Ethyl 2-Hydroxy-4-(3-methoxyphenyl)-2-methyl-5-oxo-2,5-dihydro-1H-pyrrole-3-carboxylate (**2i**).* White solid, 47 mg, 81%. R_f (acetone/DCM: 2/8) = 0.43. Mp 132−134 °C. $^{13}C\{^1H\}$ NMR (100 MHz, δ ppm/$CDCl_3$): 169.1, 163.9, 159.2, 144.3, 139.1, 130.4,

https://doi.org/10.1021/acs.joc.1c00109
J. Org. Chem. 2021, 86, 5630−5638

129.1, 122.1, 115.7, 115.0, 86.7, 61.9, 55.4, 25.5, 13.9. ^{1}H NMR (400 MHz, δ ppm/$CDCl_3$): 7.24−7.21 (m, 1H), 7.00−6.98 (m, 2H), 6.96 (s, 1H), 6.91−6.88 (m, 1H), 4.20 (q, J = 7.2 Hz, 2H), 3.75 (s, 3H), 1.75 (s, 3H), 1.13 (t, J = 7.0 Hz, 3H). HRMS (ESI) m/z: [M + Na]$^+$calcd for $C_{15}H_{17}NO_5$ Na314.0999; found 314.0997.

*Ethyl 4-(3-Bromophenyl)-2-hydroxy-2-methyl-5-oxo-2,5-dihydro-1H-pyrrole-3-carboxylate (**2j**).* Yellow solid (50 mg, 74%). R_f (acetone/DCM: 2/8) = 0.46. Mp 87−89 °C. ^{13}C{^{1}H} NMR (100 MHz, δ ppm/$CDCl_3$): 168.6, 163.4, 144.6, 138.2, 132.7, 132.7, 131.1, 129.6, 128.4, 121.9, 85.7, 62.1, 25.6, 13.9. ^{1}H NMR (400 MHz, δ ppm/$CDCl_3$): 7.60 (t, J = 1.8 Hz, 1H), 7.52−7.49 (m, 1H), 7.41−7.38 (m, 1H), 7.24−7.22 (m, 1H), 6.95 (s, 1H), 4.23 (q, J = 7.2 Hz, 2H), 3.79 (s, 1H), 1.78 (s, 3H), 1.18 (t, J = 7.2 Hz, 3H). HRMS (ESI) m/z: [M + H]$^+$calcd for $C_{14}H_{15}BrNO_4$ 340.0179; found 340.0164.

*Ethyl 4-(3-Fluorophenyl)-2-hydroxy-2-methyl-5-oxo-2,5-dihydro-1H-pyrrole-3-carboxylate (**2k**).* White solid, 34 mg, 61%; R_f (EtOAc/hexane: 3/7) = 0.44. Mp 108−110 °C. ^{13}C{^{1}H} NMR (100 MHz, δ ppm/$CDCl_3$): 168.8, 163.5, 162.3 (C−F, $^1J_{C-F}$ = 242.5 Hz,), 144.9, 138.1, 131.1 (C−F, $^3J_{C-F}$ = 8.6 Hz,), 129.6 (C−F, $^3J_{C-F}$ = 8.0 Hz,), 125.5 (C−F, $^4J_{C-F}$ = 3.2 Hz,), 116.9 (C−F, $^2J_{C-F}$ = 19.6 Hz,), 116.7 (C−F, $^2J_{C-F}$ = 17.9 Hz,), 85.8, 62.0, 25.4, 13.8. ^{1}H NMR (400 MHz, δ ppm/$CDCl_3$): 7.37−7.32 (m, 1H), 7.25−7.19 (m, 3H), 7.12−7.07 (m, 1H), 4.25 (q, J = 7.2 Hz, 2H), 3.98 (s, 1H), 1.80 (s, 3H), 1.19 (t, J = 7.2 Hz, 3H). HRMS (ESI) m/z: [M + H]$^+$ calcd for $C_{14}H_{15}FNO_4$ 280.0980; found 280.0986.

*Ethyl 4-(3,4-Dimethoxyphenyl)-2-hydroxy-2-methyl-5-oxo-2,5-dihydro-1H-pyrrole-3-carboxylate (**2l**).* Red solid, 46 mg, 71%; R_f (acetone/DCM: 2/8) = 0.35. Mp 122−124 °C. ^{13}C{^{1}H} NMR (100 MHz, δ ppm/$CDCl_3$): 169.3, 164.2, 150.6, 148.5, 142.5, 138.7, 123.3, 121.7, 113.1, 110.7, 85.5, 61.8, 56.1, 56.0, 25.6, 14.1. ^{1}H NMR (400 MHz, δ ppm/$CDCl_3$): 7.15−7.10 (m, 2H), 6.91−6.86 (m, 2H), 4.27 (q, J = 7.0 Hz, 2H), 3.90 (s, 3H), 3.87 (s, 3H), 3.71 (s, 1H), 1.80 (s, 3H), 1.22 (t, J = 7.0 Hz, 3H). HRMS (ESI) m/z: [M − H]$^-$ calcd for $C_{16}H_{18}NO_6$ 320.1140; found 320.1141.

*Ethyl 2-Hydroxy-2-methyl-5-oxo-4-(3,4,5-trimethoxyphenyl)-2,5-dihydro-1H-pyrrole-3-carboxylate (**2m**).* Orange solid, 58 mg, 82%; R_f (acetone/DCM: 2/8) = 0.29. Mp 134−136 °C. ^{13}C{^{1}H} NMR (100 MHz, δ ppm/$CDCl_3$): 168.9, 164.0, 152.9, 152.9, 143.6, 139.7, 138.7, 124.4, 107.3, 85.5, 62.0, 61.0, 56.3, 25.6, 14.1. ^{1}H NMR (400 MHz, δ ppm/$CDCl_3$): 6.75 (s, 2H), 6.69 (s, 1H), 4.28 (q, J = 7.2 Hz, 2H), 3.87 (s, 3H), 3.85 (s, 6H), 3.50 (s, 1H), 1.81 (s, 3H), 1.22 (t, J = 7.2 Hz, 3H). HRMS (ESI) m/z: [M + Na]$^+$ calcd for $C_{17}H_{21}NO_7Na$ 374.1210; found 374.1216.

*Ethyl 2-Hydroxy-2-methyl-4-(naphthalen-2-yl)-5-oxo-2,5-dihydro-1H-pyrrole-3-carboxylate (**2n**).* Orange solid, 55 mg, 88%; R_f (acetone/DCM: 2/8) = 0.45. Mp 123−125 °C. ^{13}C{^{1}H} NMR (100 MHz, δ ppm/$CDCl_3$): 169.2, 163.9, 143.9, 139.5, 133.8, 132.8, 130.1, 128.8, 127.8, 127.5, 127.2, 126.8, 126.6, 126.4, 85.7, 61.9, 25.6, 14.0. ^{1}H NMR (400 MHz, δ ppm/$CDCl_3$): 8.08 (s, 1H), 7.85−7.81 (m, 3H), 7.54−7.46 (m, 3H), 7.01 (s, 1H), 4.26 (q, J = 7.0 Hz, 2H), 3.82 (s, 1H), 1.86 (s, 3H), 1.15 (t, J = 7.0 Hz, 3H). HRMS (ESI) m/z: [M + H]$^+$ calcd for $C_{18}H_{18}NO_4$ 312.1230; found 312.1224.

*Ethyl 2-Hydroxy-2-methyl-5-oxo-4-(thiophen-2-yl)-2,5-dihydro-1H-pyrrole-3-carboxylate (**2o**).* Brown solid, 33 mg, 61%; R_f (acetone/DCM: 2/8) = 0.46. Mp 103−105 °C. ^{13}C{^{1}H} NMR (100 MHz, δ ppm/$CDCl_3$): 168.8, 163.9, 147.1, 142.6, 139.5, 131.8, 114.8, 110.7, 85.9, 62.0, 25.8, 14.2. ^{1}H NMR (400 MHz, δ ppm/$CDCl_3$): 8.40 (s, 1H), 7.43 (s, 1H), 6.91 (s, 1H), 6.87 (s, 1H), 4.40 (q, J = 7.2 Hz, 2H), 3.74 (s, 1H), 1.78 (s, 3H), 1.39 (t, J = 7.0 Hz, 3H). HRMS (ESI) m/z: [M + H]$^+$ calcd for $C_{12}H_{14}NO_4S$ 268.0638; found 268.0637.

*Ethyl 4-(Furan-2-yl)-2-hydroxy-2-methyl-5-oxo-2,5-dihydro-1H-pyrrole-3-carboxylate (**2p**).* White solid, 33 mg, 65%; R_f (acetone/DCM: 2/8) = 0.45. Mp 127−129 °C. ^{13}C{^{1}H} NMR (100 MHz, δ ppm/$CDCl_3$): 167.5, 164.4, 144.9, 144.8, 139.2, 124.8, 116.2, 112.2, 86.0, 62.0, 25.2, 14.3. ^{1}H NMR (400 MHz, δ ppm/$CDCl_3$): 7.49 (s, 1H), 7.36 (d, J = 2.8 Hz, 1H), 6.87 (s, 1H), 6.47 (s, 1H), 4.41 (q, J = 6.6 Hz, 2H), 3.87 (s, 1H), 1.74 (s, 3H), 1.37 (t, J = 7.0 Hz, 3H). HRMS (ESI) m/z: [M + H]$^+$ calcd for $C_{12}H_{14}NO_5$ 252.0866; found 252.0869.

*Ethyl 4-(Furan-3-yl)-2-hydroxy-2-methyl-5-oxo-2,5-dihydro-1H-pyrrole-3-carboxylate (**2q**).* White solid, 32 mg, 64%; R_f (acetone/DCM: 2/8) = 0.39. Mp 138−140 °C. ^{13}C{^{1}H} NMR (100 MHz, δ ppm/$CDCl_3$): 168.5, 164.1, 138.8, 132.9, 131.3, 131.2, 130.0, 127.2, 85.7, 62.1, 25.7, 14.2. ^{1}H NMR (400 MHz, δ ppm/$CDCl_3$): 8.03 (d, J = 4.4 Hz, 1H), 7.53 (d, J = 4.8 Hz, 1H), 7.10−7.08 (m, 2H), 4.10 (q, J = 7.0 Hz, 2H), 3.94 (s, 1H), 1.77 (s, 3H), 1.38 (t, J = 7.0 Hz, 3H). HRMS (ESI) m/z: [M − H]$^-$ calcd for $C_{12}H_{12}NO_5$ 250.0721; found 250.0723.

*Methyl 2-Hydroxy-2-methyl-5-oxo-4-phenyl-2,5-dihydro-1H-pyrrole-3-carboxylate (**2r**).* Brown solid, 35 mg, 71%; R_f (acetone/DCM: 2/8) = 0.47. Mp 127−129 °C. ^{13}C{^{1}H} NMR (100 MHz, δ ppm/$CDCl_3$): 169.2, 164.2, 143.8, 139.7, 129.8, 129.6, 129.2, 128.1, 85.8, 52.4, 25.4. ^{1}H NMR (400 MHz, δ ppm/$CDCl_3$): 7.47−7.44 (m, 2H), 7.40−7.35 (m, 3H), 7.27 (s, 1H), 4.05 (s, 1H), 3.74 (s, 3H), 1.78 (s, 3H). HRMS (ESI) m/z: [M + H]$^+$ calcd for $C_{13}H_{14}NO_4$ 248.0917; found 248.0911.

*Benzyl 2-Hydroxy-2-methyl-5-oxo-4-phenyl-2,5-dihydro-1H-pyrrole-3-carboxylate (**2s**).* Brown solid, 44 mg, 68%; R_f (EtOAc/hexane: 3/7) = 0.44. Mp 168−170 °C. ^{13}C{^{1}H} NMR (100 MHz, δ ppm/$CDCl_3$): 169.2, 163.6, 143.9, 139.9, 134.6, 129.6, 129.2, 128.6, 128.6, 128.5, 128.1, 85.8, 67.4, 25.4. ^{1}H NMR (400 MHz, δ ppm/$CDCl_3$): 7.32−7.26 (m, 3H), 7.23−7.18 (m, 6H), 7.04 (s, 1H), 7.02 (s, 1H), 5.12 (s, 2H), 1.71 (s, 3H). HRMS (ESI) m/z: [M + H]$^+$ calcd for $C_{19}H_{18}NO_4$ 324.1230; found 324.1225.

*tert-Butyl 2-Hydroxy-2-methyl-5-oxo-4-phenyl-2,5-dihydro-1H-pyrrole-3-carboxylate (**2t**).* White solid, 39 mg, 67%; R_f (acetone/DCM: 2/8) = 0.47. Mp 168−170 °C. ^{13}C{^{1}H} NMR (100 MHz, δ ppm/$CDCl_3$): 169.3, 163.2, 145.2, 138.7, 129.7, 129.5, 128.0, 85.6, 83.8, 28.0, 25.6. ^{1}H NMR (400 MHz, δ ppm/$CDCl_3$): 7.48−7.46 (m, 2H), 7.40−7.38 (m, 3H), 6.70 (s, 1H), 1.80 (s, 3H), 1.38 (s, 9H). HRMS (ESI) m/z: [M + H]$^+$ calcd for $C_{16}H_{20}NO_4$ 290.1387; found 290.1382.

General Procedure for the Heteroarylation of 2-Hydroxy-2,5-dihydro-1*H*-pyrrole-3-carboxylates 4a−h. DCM (2.0 mL) was added to a mixture of 2-hydroxy-2,5-dihydro-1*H*-pyrrole-3-carboxylate **2** (0.20 mmol), indole **3** (0.26 mmol), PPh_3AuCl (0.02 mmol), and AgOTf (0.02 mmol) under an air atmosphere, and the resulting mixture was stirred at room temperature for 30 min. After the completion of the reaction, as indicated by TLC, the solvent was evaporated off, followed by extraction using ethyl acetate and water. The organic layer was dried over Na_2SO_4 and evaporated under reduced pressure. The residue was subjected to column chromatography (eluent: ethyl acetate/hexane = 40/60) to afford the product **4**.

*Ethyl 2-Methyl-2-(1-methyl-indol-3-yl)-5-oxo-4-phenyl-2,5-dihydro-1H-pyrrole-3-carboxylate (**4a**).* White solid, 64 mg, 86%; R_f (EtOAc/hexane: 3/7) = 0.37. Mp 158−160 °C. ^{13}C{^{1}H} NMR (100 MHz, δ ppm/$CDCl_3$): 169.9, 163.6, 148.4, 139.2, 137.6, 130.6, 129.5, 129.5, 129.2, 128.0, 125.5, 122.0, 120.1, 119.9, 112.4, 109.6, 61.4, 61.0, 33.1, 26.1 13.6. ^{1}H NMR (400 MHz, δ ppm/$CDCl_3$): 7.57−7.55 (m, 2H), 7.49 (d, J = 8.0 Hz, 1H), 7.42−7.40 (m, 3H), 7.30−7.19 (m, 2H), 7.13 (s, 1H), 7.08−7.04 (m, 1H), 6.52 (s, 1H), 4.01−3.89 (m, 2H), 3.77 (s, 3H), 2.13 (s, 3H), 0.88 (t, J = 7.0 Hz, 3H). HRMS (ESI) m/z: [M + H]$^+$ calcd for $C_{23}H_{23}N_2O_3$ 375.1703; found 375.1704.

*Ethyl 2-(1H-Indol-3-yl)-2-methyl-5-oxo-4-phenyl-2,5-dihydro-1H-pyrrole-3-carboxylate (**4b**).* White solid, 56 mg, 77%; R_f (EtOAc/hexane: 4/6) = 0.20. Mp 158−160 °C. ^{13}C{^{1}H} NMR (100 MHz, δ ppm/DMSO-d_6): 168.6, 163.4, 148.3, 137.6, 136.7, 130.8, 129.0, 129.0, 128.7, 127.8, 124.7, 124.2, 121.0, 119.0, 118.8, 112.7, 111.7, 60.7, 60.4, 25.6, 13.3. ^{1}H NMR (400 MHz, δ ppm/DMSO-d_6): 11.11 (s, 1H), 9.22 (s, 1H), 7.56−7.54 (m, 2H), 7.46−7.39 (m, 5H), 7.34 (d, J = 4.0 Hz, 1H), 7.09 (t, J = 7.4 Hz, 1H), 6.97 (t, J = 7.4 Hz, 1H), 4.00−3.86 (m, 2H), 2.05 (s, 3H), 0.86 (t, J = 7.0 Hz, 3H). HRMS (ESI) m/z: [M + H]$^+$ calcd for $C_{22}H_{21}N_2O_3$ 361.1547; found 361.1539.

*Ethyl 2-(5-Bromo-1H-indol-3-yl)-2-methyl-5-oxo-4-phenyl-2,5-dihydro-1H-pyrrole-3-carboxylate (**4c**).* Yellow solid, 59 mg, 67%; R_f (EtOAc/hexane: 4/6) = 0.20. Mp 226−228 °C. ^{13}C{^{1}H} NMR

https://doi.org/10.1021/acs.joc.1c00109
J. Org. Chem. 2021, 86, 5630−5638

(100 MHz, δ ppm/DMSO-d_6): 168.5, 163.2, 148.1, 137.6, 135.4, 130.6, 128.9, 128.8, 127.8, 126.4, 125.9, 123.5, 121.2, 113.8, 112.6, 111.3, 60.5, 60.4, 25.4, 13.3. ^{1}H NMR (400 MHz, δ ppm/DMSO-d_6): 11.34 (s, 1H), 9.26 (s, 1H), 7.53−7.43 (m, 6H), 7.42−7.41 (m, 1H), 7.38−7.36 (m, 1H), 7.22−7.19 (m, 1H), 4.02−3.89 (m, 2H), 2.01 (s, 3H), 0.86 (t, J = 7.0 Hz, 3H). HRMS (ESI) m/z: $[M + H]^+$ calcd for $C_{22}H_{20}BrN_2O_3$ 439.0652; found 439.0658.

*Ethyl 2-(6-Bromo-1H-indol-3-yl)-2-methyl-5-oxo-4-phenyl-2,5-dihydro-1H-pyrrole-3-carboxylate (**4d**).* White solid, 58 mg, 66%; R_f (EtOAc/hexane: 4/6) = 0.20. Mp 208−210 °C. $^{13}C\{^1H\}$ NMR (100 MHz, δ ppm/DMSO-d_6)): 168.5, 163.3, 148.1, 137.6, 130.6, 129.0, 129.0, 128.7, 127.7, 125.4, 123.7, 121.7, 120.5, 114.3, 113.8, 113.1, 60.4, 60.4, 25.4, 13.3. ^{1}H NMR (400 MHz, δ ppm/DMSO-d_6): 11.25 (s, 1H), 9.23 (s, 1H), 7.57 (s, 1H), 7.50−7.43 (m, 6H), 7.22−7.20 (m, 1H), 7.12−7.10 (m, 1H), 4.03−3.89 (m, 2H), 2.00 (s, 3H), 0.84 (t, J = 7.0 Hz, 3H). HRMS (ESI) m/z: $[M + H]^+$ calcd for $C_{22}H_{20}BrN_2O_3$ 439.0652; found 439.0655.

*Ethyl 2-Methyl-5-oxo-4-phenyl-2-(1,6,7,8-tetrahydrocyclopenta-[g]indol-3-yl)-2,5-dihydro-1H-pyrrole-3-carboxylate (**4e**).* White solid, 53 mg, 66%; R_f (EtOAc/hexane: 4/6) = 0.29. Mp 238−240 °C. $^{13}C\{^1H\}$ NMR (100 MHz, δ ppm/DMSO-d_6): 168.5, 163.3, 148.4, 137.6, 136.5, 133.8, 130.8, 129.0, 128.6, 127.7, 125.6, 123.4, 123.3, 117.0, 115.7, 113.2, 60.7, 60.3, 32.5, 29.9, 25.5, 24.8, 13.3. ^{1}H NMR (400 MHz, δ ppm/DMSO-d_6): 10.98 (s, 1H), 9.17 (s, 1H), 7.50−7.43 (m, 5H), 7.34 (s, 1H), 7.10 (d, J = 7.6 Hz, 1H), 6.86 (d, J = 8.0 Hz, 1H), 3.98−3.85 (m, 2H), 3.02 (t, J = 6.8 Hz, 2H), 2.93 (t, J = 6.2 Hz, 2H), 2.13−2.09 (m, 2H), 2.03 (s, 3H), 0.88 (t, J = 6.8 Hz, 3H). HRMS (ESI) m/z: $[M + H]^+$ calcd for $C_{25}H_{25}N_2O_3$ 401.1860; found 401.1865.

*Ethyl 2-Methyl-2-(3-methyl-1H-indol-2-yl)-5-oxo-4-phenyl-2,5-dihydro-1H-pyrrole-3-carboxylate (**4f**).* White solid, 60 mg, 80%; R_f (EtOAc/hexane: 4/6) = 0.57. Mp 87−90 °C. $^{13}C\{^1H\}$ NMR (100 MHz, δ ppm/DMSO-d_6): 168.6, 162.9, 145.9, 138.8, 134.8, 130.5, 130.4, 129.3, 129.0, 128.8, 127.6, 121.0, 118.3, 117.7, 111.1, 106.5, 60.9, 60.6, 25.0, 13.3, 8.7. ^{1}H NMR (400 MHz, δ ppm/DMSO-d_6): 10.80 (s, 1H), 9.35 (s, 1H), 7.49−7.42 (m, 6H), 7.36−7.34 (m, 1H), 7.06 (t, J = 6.0 Hz, 1H), 6.96 (t, J = 5.8 Hz, 1H), 3.98−3.90 (m, 2H), 2.15 (s, 3H), 2.07 (s, 3H), 0.85 (t, J = 5.6 Hz, 3H). HRMS (ESI) m/z: $[M + H]^+$ calcd for $C_{23}H_{23}N_2O_3$ 375.1703; found 375.1693.

*Ethyl 2-(5-Methoxythiophen-2-yl)-2-methyl-5-oxo-4-phenyl-2,5-dihydro-1H-pyrrole-3-carboxylate (**4g**).* Orange solid, 51 mg, 72%; R_f (EtOAc/hexane: 4/6) = 0.50. Mp 125−128 °C. $^{13}C\{^1H\}$ NMR (100 MHz, δ ppm/$CDCl_3$): 169.5, 166.3, 163.3, 148.0, 138.6, 130.1, 129.5, 129.4, 128.0, 123.1, 103.7, 62.4, 61.3, 60.4, 25.6, 13.8. ^{1}H NMR (400 MHz, δ ppm/$CDCl_3$): 7.51−7.49 (m, 2H), 7.40−7.38 (m, 3H), 6.71 (d, J = 4.0 Hz, 1H), 6.63 (s, 1H), 6.03 (d, J = 3.6 Hz, 1H), 4.14−4.06 (m, 2H), 3.84 (s, 3H), 2.05 (s, 3H), 1.04 (t, J = 7.2 Hz, 3H). HRMS (ESI) m/z: $[M + H]^+$ calcd for $C_{19}H_{20}NO_4$ S358.1108; found 358.1109.

*Ethyl 4-(4-Methoxyphenyl)-1′,2-dimethyl-5-oxo-2,5-dihydro-1H,1′H-[2,2′-bipyrrole]-3-carboxylate (**4h**).* White solid, 48 mg, 67%; R_f (EtOAc/hexane: 4/6)= 0.22. Mp 158−160 °C. $^{13}C\{^1H\}$ NMR (100 MHz, δ ppm/$CDCl_3$): 170.2, 164.0, 160.8, 146.5, 136.9, 130.7, 130.7, 127.0, 125.7, 122.2, 113.8, 110.6, 106.5, 61.4, 60.6, 55.5, 34.8, 27.2, 13.9. ^{1}H NMR (400 MHz, δ ppm/$CDCl_3$): 7.59−7.55 (m, 2H), 6.96−6.92 (m, 2H), 6.55−6.54 (m, 1H), 6.41 (s, 1H), 6.28−6.26 (m, 1H), 6.03−6.01 (m, 1H), 4.09 (q, J = 7.0 Hz, 2H), 3.84 (s, 3H), 3.47 (s, 3H), 2.03 (s, 3H), 1.04 (t, J = 7.0 Hz, 3H). HRMS (ESI) m/z: $[M + H]^+$ calcd for $C_{20}H_{23}N_2O_4$ 355.1652; found 355.1647.

Synthesis of Boc-Protected 2-Hydroxy-2-methyl-5-oxo-4-phenyl-2,5-dihydro-1*H*-pyrrole-3-carboxylate 5. To a mixture of 2-hydroxy-2-methyl-5-oxo-4-phenyl-2,5-dihydro-1*H*-pyrrole-3-carboxylate **4b** (0.4 mmol, 144 mg), triethylamine (0.4 mmol, 40 mg) and DMAP (0.2 mmol, 24 mg) in THF (3 mL) was added $(Boc)_2O$ (1.0 mmol, 218 mg) slowly at 0 °C. The reaction mixture was warmed up to 25 °C and stirred for 8 h. After the completion of the reaction, as indicated by TLC, the solvent was evaporated off, followed by extraction using ethyl acetate and water. The organic layer was dried over Na_2SO_4 and evaporated under reduced pressure. The residue was subjected to column chromatography (eluent: ethyl acetate/hexane = 20/80) to afford the product **5** as white solid (122 mg, 55%). R_f (EtOAc/hexane: 2/8) = 0.38. Mp 120−122 °C. $^{13}C\{^1H\}$ NMR (100 MHz, δ ppm/$CDCl_3$): 167.3, 163.3, 149.6, 148.8, 148.7, 137.4, 135.8, 129.7, 129.6, 129.3, 128.2, 127.5, 125.4, 124.6, 123.3, 119.3, 118.0, 115.5, 84.3, 83.5, 65.3, 61.5, 28.3, 27.9, 24.3, 13.5. ^{1}H NMR (400 MHz, δ ppm/$CDCl_3$): 8.15 (d, J = 8.4 Hz, 1H), 7.66 (s, 1H), 7.53−7.51 (m, 2H), 7.42−7.40 (m, 3H), 7.29−7.27 (m, 1H), 7.25−7.23 (m, 1H), 7.16−7.12 (m, 1H), 3.99−3.93 (m, 2H), 2.30 (s, 3H), 1.68 (s, 9H), 1.24 (s, 9H), 0.86 (t, J = 7.0 Hz, 3H). HRMS (ESI) m/z: $[M + H]^+$ calcd for $C_{32}H_{37}N_2O_7$ 561.2595; found 561.2604.

Synthesis of 1-(*tert*-Butyl)-3-ethyl-2-(1-(*tert*-butoxycarbonyl)-1*H*-indol-3-yl)-5-hydroxy-2-methyl-4-phenyl-2,5-dihydro-1*H*-pyrrole-1,3-dicarboxylate 6. A solution of DIBAL-H (0.8 mL, 1.0 M in hexane) was added to a solution of 1-(*tert*-butyl)-3-ethyl-2-(1-(*tert*-butoxycarbonyl)-1*H*-indol-3-yl)-2-methyl-5-oxo-4-phenyl-2,5-dihydro-1*H* pyrrole-1,3-dicarboxylate **5** (112 mg, 0.20 mmol) in THF (3 mL) under N_2 at −78 °C. The reaction mixture was stirred for 30 min at the same temperature. After the completion of the reaction, as indicated by TLC, the solvent was evaporated off, followed by extraction using ethyl acetate and water. The organic layer was dried over Na_2SO_4 and evaporated under reduced pressure. The residue was subjected to column chromatography (eluent: ethyl acetate/hexane = 15/85) to afford the product **6** as white solid (82 mg, 73%). Diastereomeric ratio: 5:1. R_f (EtOAc/hexane: 2/8) = 0.47. Mp 156−158 °C. $^{13}C\{^1H\}$ NMR (100 MHz, δ ppm/$CDCl_3$): 163.9, 154.3, 149.9, 142.9, 135.8, 135.5, 132.9, 128.9, 128.7, 128.5, 128.3, 124.6, 124.1, 122.8, 121.9, 121.3, 114.9, 87.2, 83.7, 81.2, 67.5, 60.7, 28.3, 28.1, 25.8, 13.4. ^{1}H NMR (400 MHz, δ ppm/$CDCl_3$): 8.23 (d, J = 7.6 Hz, 1H), 8.10 (d, J = 8.0 Hz, 1H), 7.54−7.47 (m, 3H), 7.36−7.35 (m, 3H), 7.24−7.21 (m, 1H), 7.18−7.14 (m, 1H), 6.46 (d, J = 2.4 Hz, 1H), 4.76 (d, J = 2.8 Hz, 1H), 3.87−3.79 (m, 2H), 2.11 (s, 3H), 1.65 (s, 9H), 1.07 (s, 9H), 0.77 (t, J = 7.2 Hz, 3H). HRMS (ESI) m/z: $[M - H]^-$ calcd for $C_{32}H_{37}N_2O_7$ 561.2606; found 561.2614.

Synthesis of 1-(*tert*-Butyl)-3-ethyl-2-(1-(*tert*-butoxycarbonyl)-1*H*-indol-3-yl)-2-methyl-5-(1-methyl-1*H*-indol-3-yl)-4-phenyl-2,5-dihydro-1*H*-pyrrole-1,3-dicarboxylate 7. DCM (2.0 mL) was added to a mixture of 1-(*tert*-butyl)-3-ethyl-2-(1-(*tert*-butoxycarbonyl)-1*H*-indol-3-yl)-5-hydroxy-2-methyl-4-phenyl-2,5-dihydro-1*H*-pyrrole-1,3-dicarboxylate **6** (112 mg, 0.20 mmol), 1-methylindole **3a** (39 mg, 0.30 mmol), and $In(OTf)_3$ (11 mg, 0.02 mmol) under an air atmosphere. The resulting mixture was stirred at 25 °C for 15 min. After the completion of the reaction, as indicated by TLC, the solvent was evaporated off, followed by extraction using ethyl acetate and water. The organic layer was dried over Na_2SO_4 and evaporated under reduced pressure. The residue was subjected to column chromatography (eluent: ethyl acetate/hexane = 20/80) to afford the product **7** as white solid (90 mg, 67%). Diastereomeric ratio: 3:1. R_f (EtOAc/hexane: 2/8) = 0.32. Mp 162−164 °C. $^{13}C\{^1H\}$ NMR (100 MHz, δ ppm/$CDCl_3$): 164.5, 153.1, 150.0, 145.8, 137.3, 135.9, 133.9, 132.3, 131.0, 128.4, 128.0, 127.9, 126.8, 124.3, 124.2, 123.2, 121.2, 120.5, 119.6, 119.3, 115.3, 111.9, 109.4, 83.7, 79.6, 68.4, 64.3, 60.4, 33.0, 28.4, 27.9, 25.3, 13.4. ^{1}H NMR (400 MHz, δ ppm/$CDCl_3$): 8.23 (s, 1H), 7.91 (s, 1H), 7.64−7.56 (m, 2H), 7.36−7.18 (m, 11H), 6.69 (s, 1H), 3.93−3.89 (m, 2H), 3.77 (s, 3H), 2.38 (s, 3H), 1.72 (s, 9H), 0.88 (s, 9H), 0.83−0.79 (m, 3H). HRMS (ESI) m/z: $[M + H]^+$ calcd for $C_{41}H_{46}N_3O_6$ 676.3381; found 676.3382.

Synthesis of Ethyl-2-methyl-5-oxo-4-phenyl-2,5-dihydro-1*H*-pyrrole-3-carboxylate 8. To a stirred solution of ethyl-2-hydroxy-2-methyl-5-oxo-4-phenyl-2,5-dihydro-1*H*-pyrrole-3-carboxylate **2a** (52 mg, 0.20 mmol) in 2 mL of THF, $NaBH_3CN$ (38 mg, 0.60 mmol) was added in one portion at 25 °C. After 5 min, the reaction mixture was acidified with 1.5 equiv of 35% HCl and stirred for 5 h. After the completion of the reaction, as indicated by TLC, the solvent was evaporated off, followed by extraction using ethyl acetate and water. The organic layer was dried over Na_2SO_4 and evaporated under reduced pressure. The residue was subjected to a column chromatography (eluent: ethyl acetate/hexane = 30/70) to afford the product **8** as yellow solid (27 mg, 56%). R_f (EtOAc/hexane: 4/6) = 0.42. Mp 101−103 °C. $^{13}C\{^1H\}$ NMR (100 MHz, δ ppm/$CDCl_3$):

https://doi.org/10.1021/acs.joc.1c00109
J. Org. Chem. 2021, 86, 5630−5638

179.7, 164.0, 152.3, 136.2, 128.8, 128.8, 128.1, 128.1, 127.6, 109.5, 59.8, 54.1, 14.2, 13.6. ^{1}H NMR (400 MHz, δ ppm/$CDCl_3$): 8.75 (s, 1H), 7.35−7.31 (m, 2H), 7.29−7.27 (m, 1H), 7.22−7.19 (m, 2H), 4.40 (q, J = 2.2 Hz, 1H), 4.11−3.99 (m, 2H), 2.41 (d, J = 2.4 Hz, 3H), 1.08 (t, J = 7.2 Hz, 3H). HRMS (ESI) m/z: $[M + H]^+$ calcd for $C_{14}H_{16}NO_3$ 246.1125; found 246.1124.

Synthesis of Ethyl-2-methoxy-2-methyl-5-oxo-4-phenyl-2,5-dihydro-1*H*-pyrrole-3-carboxylate 9. To an oven-dried round-bottom flask was added ethyl-2-hydroxy-2-methyl-5-oxo-4-phenyl-2,5-dihydro-1*H*-pyrrole-3-carboxylate **2a** (52 mg, 0.20 mmol) in MeOH (2 mL). The reaction mixture was acidified with 1 equiv of 35% HCl and stirred for 1 h at 25 °C. After the completion of the reaction, as indicated by TLC, the solvent was evaporated off, followed by extraction using ethyl acetate and water. The organic layer was dried over Na_2SO_4 and evaporated under reduced pressure. The residue was subjected to column chromatography (eluent: ethyl acetate/hexane = 30/70) to afford the product **9** as yellow solid (39 mg, 70%). R_f (EtOAc/hexane: 4/6) = 0.32. Mp 97−99 °C. $^{13}C\{^1H\}$ NMR (100 MHz, δ ppm/$CDCl_3$): 169.1, 163.5, 143.6, 140.0, 129.8, 129.4, 129.2, 128.3, 89.5, 61.7, 50.7, 25.1, 14.0. ^{1}H NMR (400 MHz, δ ppm/$CDCl_3$): 7.55−7.52 (m, 2H), 7.43−7.39 (m, 3H), 6.27 (s, 1H), 4.29−4.22 (m, 2H), 3.23 (s, 3H), 1.81 (s, 3H), 1.19 (t, J = 7.4 Hz, 3H). HRMS (ESI) m/z: $[M + H]^+$ calcd for $C_{15}H_{18}NO_4$ 276.1230; found 276.1228.

(*Z*)-Ethyl-5-(hydroxyimino)-2-methyl-4-phenyl-4,5-dihydrofuran-3-carboxylate (Intermediate F). Following the Kwons procedure,[14] treatment of ethyl-2-(2-nitro-1-phenylethyl)buta-2,3-dienoate **1a** (52 mg, 0.20 mmol) with PBu_3 (2 mg, 0.01 mmol) in CH_3CN (2 mL) at 25 °C for 40 min followed by column chromatography afforded the intermediate **F** as colorless oil (36 mg, 70%). R_f (EtOAc/hexane: 3/7) = 0.15. $^{13}C\{^1H\}$ NMR (100 MHz, δ ppm/$CDCl_3$): 164.3, 163.5, 158.8, 138.7, 128.8, 127.8, 110.4, 60.4, 49.2, 14.07, 13.8. ^{1}H NMR (400 MHz, δppm/$CDCl_3$): 7.32−7.24 (m, 3H), 7.20−7.17 (m, 2H), 7.08 (s, 1H), 4.88 (q, J = 2.0 Hz, 1H), 4.11−3.99 (m, 2H), 2.49 (d, J = 2.0 Hz, 3H), 1.08 (t, J = 7.2 Hz, 3H). HRMS (ESI) m/z: $[M + H]^+$ calcd for $C_{14}H_{16}NO_4$ 262.1074; found 262.1076.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.joc.1c00109.

Copies of NMR spectra and CIFs for **2f** and **4e** (PDF)

FAIR data, including the primary NMR FID files, for compounds **2a−t**, **4a−h**, **5−9**, Int-**F** (ZIP)

Accession Codes

CCDC 2006864 and 2009054 contain the supplementary crystallographic data for this paper. These data can be obtained free of charge via www.ccdc.cam.ac.uk/data_request/cif, or by emailing data_request@ccdc.cam.ac.uk, or by contacting The Cambridge Crystallographic Data Centre, 12 Union Road, Cambridge CB2 1EZ, UK; fax: + 44 1223 336033.

■ AUTHOR INFORMATION

Corresponding Author

Kishor Mohanan − *Medicinal and Process Chemistry Division, CSIR-Central Drug Research Institute, Lucknow 226031, India; Academy of Scientific and Innovative Research, Ghaziabad 201002, India;* orcid.org/0000-0001-5602-7112; Email: kishor.mohanan@cdri.res.in

Authors

Narendra Kumar Vaishanv − *Medicinal and Process Chemistry Division, CSIR-Central Drug Research Institute, Lucknow 226031, India*

Mohd Khalid Zaheer − *Medicinal and Process Chemistry Division, CSIR-Central Drug Research Institute, Lucknow 226031, India*

Sandeep Kumar − *Medicinal and Process Chemistry Division, CSIR-Central Drug Research Institute, Lucknow 226031, India*

Ruchir Kant − *Molecular and Structural Biology Division, CSIR-Central Drug Research Institute, Lucknow 226031, India*

Complete contact information is available at: https://pubs.acs.org/10.1021/acs.joc.1c00109

Notes

The authors declare no competing financial interest.

■ ACKNOWLEDGMENTS

N.K.V. and M.K.Z. thank UGC-New Delhi for Research Fellowship. S.K. is grateful to CSIR-New Delhi for the award of Junior Research Fellowship. We thank Dr. Tejender S. Thakur, MSB division, CSIR-CDRI for supervising the X-ray data collection and structure determination. We thank the SAIF division for analytical support. CDRI Communication No. 10217.

■ REFERENCES

(1) (a) Kakeya, H.; Onozawa, C.; Sato, M.; Arai, K.; Osada, H. Neuritogenic Effect of Epolactaene Derivatives on Human Neuroblastoma Cells Which Lack High-Affinity Nerve Growth Factor Receptors. *J. Med. Chem.* **1997**, *40*, 391−394. (b) Toja, E.; Gorini, C.; Barzaghi, F.; Galliani, G. Derivatives of 1-arylsulphonyl-1,5-dihydro-2H-pyrrol-2-one, and compositions containing them. Patent US4885307A, December 5, 1989.

(2) (a) Uesugi, S.; Fujisawa, N.; Yoshida, J.; Watanabe, M.; Dan, S.; Yamori, T.; Shiono, Y.; Kimura, K. Pyrrocidine A, a metabolite of endophytic fungi, has a potent apoptosis-inducing activity against HL60 cells through caspase activation via the Michael addition. *J. Antibiot.* **2016**, *69*, 133−140. (b) Singh, S. B.; Goetz, M. A.; Jones, E. T.; Bills, G. F.; Giacobbe, R. A.; Herranz, L.; Stevens-Miles, S.; Williams, D. L. Oteromycin: A Novel Antagonist of Endothelin Receptor. *J. Org. Chem.* **1995**, *60*, 7040−7042. (c) Kakeya, E.; Takahashi, I.; Okada, G.; Isono, K.; Osada, H. Epolactaene, a Novel Neuritogenic Compound in Human Neuroblastoma Cells, Produced by a Marine Fungus. *J. Antibiot.* **1995**, *48*, 733−735.

(3) (a) Coleman, R. S.; Walczak, M. C.; Campbell, E. L. Total Synthesis of Lucilactaene, A Cell Cycle Inhibitor Active in p53-Inactive Cells. *J. Am. Chem. Soc.* **2005**, *127*, 16038−16039. (b) Snider, B. B.; Neubert, B. J. A Novel Biomimetic Route to the 3-Acyl-5-hydroxy-3-pyrrolin-2-one and 3-Acyl-3,4-epoxy-5-hydroxypyrrolidin-2-one Ring Systems. *J. Org. Chem.* **2004**, *69*, 8952−8955.

(4) Mardjan, M. I. D.; Parrain, J.-L.; Commeiras, L. Strategies To Access γ-Hydroxy-γ-butyrolactams. *Synthesis* **2018**, *50*, 1175−1198.

(5) (a) Tang, Y.; Lv, M.; Liu, X.; Feng, H.; Liu, L. Synthesis of β-Halo-pyrrolidinones through a Tandem Sequence of 5-Endo Halolactamization and C-H Oxidative Functionalization. *Org. Lett.* **2013**, *15*, 1382−1385. (b) Ma, S.; Xie, H. Unexpected Facile Sequential Halolactamization-Hydroxylation of 2,3-Allenamides with CuX_2 for the Efficient Synthesis of 4-Halo-5-hydroxypyrrol-2(5*H*)-ones. *Org. Lett.* **2000**, *2*, 3801−3803.

(6) (a) Yang, L.; Lei, C.-H.; Wang, D.-X.; Huang, Z.-T.; Wang, M.-X. Highly Efficient and Expedient Synthesis of 5-Hydroxy-1*H*-pyrrol-2-(5*H*)-ones from $FeCl_3$-Catalyzed Tandem Intramolecular Enaminic Addition of Tertiary Enamides to Ketones and 1,3-Hydroxy Rearrangement. *Org. Lett.* **2010**, *12*, 3918−3921. (b) Clark, A. J.; Dell, C. P.; McDonagh, J. M.; Geden, J.; Mawdsley, P. Oxidative 5-Endo Cyclization of Enamides Mediated by Ceric Ammonium Nitrate. *Org. Lett.* **2003**, *5*, 2063−2066.

(7) Wu, M.-Y.; Li, K.; Wang, N.; He, T.; Yu, X.-Q. A Novel Catalyst-Free Tandem Reaction for the Synthesis of 5-Hydroxy-1,5-dihydro-2H-pyrrol-2-ones in Water Medium. *Synthesis* **2011**, *2011*, 1831−1839.
(8) Lim, C.-H.; Kim, S.-H.; Kim, J.-N. Facile Synthesis of 5-Hydroxy-3-pyrrolin-2-ones from Morita-Baylis-Hillman Adducts. *Bull. Korean Chem. Soc.* **2012**, *33*, 1622−1626.
(9) For selected reviews, see: (a) Ni, H.; Chan, W.-L.; Lu, Y. Phosphine-Catalyzed Asymmetric Organic Reactions. *Chem. Rev.* **2018**, *118*, 9344−9411. (b) Guo, H.; Fan, Y. C.; Sun, Z.; Wu, Y.; Kwon, O. Phosphine Organocatalysis. *Chem. Rev.* **2018**, *118*, 10049−10293. (c) Wang, Z.; Xu, X.; Kwon, O. Phosphine Catalysis of Allenes with Electrophiles. *Chem. Soc. Rev.* **2014**, *43*, 2927−2940. (d) Fan, Y. C.; Kwon, O. Advances in Nucleophilic Phosphine Catalysis of Alkenes, Allenes, Alkynes, and MBHADs. *Chem. Commun.* **2013**, *49*, 11588−11619. (e) Zhao, Q.-Y.; Lian, Z.; Wei, Y.; Shi, M. Development of Asymmetric Phosphine-Promoted Annulations of Allenes with Electron-Deficient Olefins and Imines. *Chem. Commun.* **2012**, *48*, 1724−1732. (f) Ye, L.-W.; Zhou, J.; Tang, Y. Phosphine-Triggered Synthesis of Functionalized Cyclic Compounds. *Chem. Soc. Rev.* **2008**, *37*, 1140−1152.
(10) (a) Zhou, L.; Wang, C.; Yuan, C.; Liu, H.; Zhang, C.; Guo, H. Phosphine-Catalyzed [3 + 2] Annulation of 2-Hydroxy-1,4-naphthaquinones and Allenoate: An Allene-Alkene [3 + 2] Annulation Mechanism Involving Consecutive γ-Addition-Aldol Reaction. *Org. Lett.* **2018**, *20*, 6591−6595. (b) Zielke, K.; Waser, M. Formal (4 + 1)-Addition of Allenoates to *o*-Quinone Methides. *Org. Lett.* **2018**, *20*, 768−771. (c) Zhou, W.; Wang, H.; Tao, M.; Zhu, C.-Z.; Lin, T.-Y.; Zhang, J. Phosphine-Catalyzed Enantioselective [3 + 2] Cycloadditions of γ-substituted Allenoates with β-perfluoroalkyl Enones. *Chem. Sci.* **2017**, *8*, 4660−4665. (d) Yao, W.; Dou, X.; Lu, Y. Highly Enantioselective Synthesis of 3,4-Dihydropyrans through a Phosphine-Catalyzed [4 + 2] Annulation of Allenones and β,γ-Unsaturated α-Keto Esters. *J. Am. Chem. Soc.* **2015**, *137*, 54−57. (e) Fujiwara, Y.; Fu, G. C. Application of a New Chiral Phosphepine to the Catalytic Asymmetric Synthesis of Highly Functionalized Cyclopentenes That Bear an Array of Heteroatom-Substituted Quaternary Stereocenters. *J. Am. Chem. Soc.* **2011**, *133*, 12293−12297. (f) Xiao, H.; Chai, Z.; Zheng, C. W.; Yang, Y. Q.; Liu, W.; Zhang, J. K.; Zhao, G. Asymmetric [3 + 2] Cycloadditions of Allenoates and Dual Activated Olefins Catalyzed by Simple Bifunctional N-Acyl Aminophosphines. *Angew. Chem., Int. Ed.* **2010**, *49*, 4467−4470. (g) Zhu, X.-F.; Lan, J.; Kwon, O. An Expedient Phosphine-Catalyzed [4 + 2] Annulation: Synthesis of Highly Functionalized Tetrahydropyridines. *J. Am. Chem. Soc.* **2003**, *125*, 4716−4717. (h) Zhang, C.; Lu, X. Phosphine-Catalyzed Cycloaddition of 2,3-Butadienoates or 2-Butynoates with Electron-Deficient Olefins. A Novel [3 + 2] Annulation Approach to Cyclopentenes. *J. Org. Chem.* **1995**, *60*, 2906−2908.
(11) (a) Zhang, Q.-Q.; Chen, S.-Y.; Lin, E.; Wang, H.; Li, Q. Regio- and Stereoselective Alkenylation of Allenoates with gem Difluoroalkenes: Facile Access to Fluorinated 1,4-Enynes Bearing an All-Carbon Quaternary Center. *Org. Lett.* **2019**, *21*, 3123−3126. (b) Vaishanv, N. K.; Gupta, A. K.; Kant, R.; Mohanan, K. Fluoride-Mediated α-Selective 1,6-Conjugate Addition of Allenic Esters to *p*-Quinone Methides. *J. Org. Chem.* **2018**, *83*, 8759−8767. (c) Wang, M.; Fang, Z.; Fu, C.; Ma, S. Metal Alkoxide Promoted Regio- and Stereoselective C = O and C = C Metathesis of Allenoates with Aldehydes. *Angew. Chem., Int. Ed.* **2014**, *53*, 3214−3217. (d) Yang, H.; Xu, B.; Hammond, G. B. Highly Regioselective Fluorination and Iodination of Alkynyl Enolates. *Org. Lett.* **2008**, *10*, 5589−5591. (e) Wang, W.; Xu, B.; Hammond, G. B. Synthesis of Functionalized α,α-Disubstituted α-Alkynyl Esters from Allenoates through an Alkynylenolate Intermediate. *Org. Lett.* **2008**, *10*, 3713−3716.
(12) Shi, M.; Dai, L.-Z.; Shi, Y.-L.; Zhao, G.-L. Potassium Carbonate-Catalyzed Reactions of Salicylic Aldehydes with Allenic Ketones and Esters: An Effective Way to Synthesize Functionalized 2*H*-Chromenes. *Adv. Synth. Catal.* **2006**, *348*, 967−972.
(13) Selig, P.; Raven, W. A Convenient Allenoate-Based Synthesis of 2-Quinolin-2-yl Malonates and β-Ketoesters. *Org. Lett.* **2014**, *16*, 5192−5195.
(14) Yao, C.; Bao, Y.; Lu, T.; Zhou, Q. Stereoselective Synthesis of Functionalized Benzooxazepino[5,4-α]isoindolone Derivatives via Cesium Carbonate Catalyzed Formal [5 + 2] Annulation of 2-(2-Hydroxyphenyl)isoindoline-1,3-dione with Allenoates. *Org. Lett.* **2018**, *20*, 2152−2155.
(15) Rouh, H.; Liu, Y.; Katakam, N.; Pham, L.; Zhu, Y.-L.; Li, G. Synthesis of Functionalized Chromene and Chroman Derivatives via Cesium Carbonate Promoted Formal [4 + 2] Annulation of 2′-Hydroxychalcones with Allenoates. *J. Org. Chem.* **2018**, *83*, 15372−15379.
(16) Zhou, Q.-F.; Zhang, K.; Cai, L.; Kwon, O. Phosphine-Catalyzed Intramolecular Cyclizations of α-Nitroethylallenoates Forming (Z)-Furanone Oximes. *Org. Lett.* **2016**, *18*, 2954−2957.
(17) Kochanowska-Karamyan, A. J.; Hamann, M. T. Marine Indole Alkaloids: Potential New Drug Leads for the Control of Depression and Anxiety. *Chem. Rev.* **2010**, *110*, 4489−4497.
(18) (a) Carbone, A.; Spanò, V.; Parrino, B.; Ciancimino, C.; Attanasi, O. A.; Favi, G. A Facile Synthesis of Deaza-Analogues of the Bisindole Marine Alkaloid Topsentin. *Molecules* **2013**, *18*, 2518−2527. (b) Oh, K.-B.; Mar, W.; Kim, S.; Kim, J.-Y.; Oh, M.-N.; Kim, J.-G.; Shin, D.; Sim, C. J.; Shin, J. Bis(indole) Alkaloids as Sortase Inhibitors from the Sponge Spongosorites sp. *Bioorg. Med. Chem. Lett.* **2005**, *15*, 4927−4931.

https://doi.org/10.1021/acs.joc.1c00109
J. Org. Chem. 2021, 86, 5630−5638